DIAGNOSTIC TEST
GENETICS

Multiple Choice Questions

1. The number of chromosomes in a human body cell is:
   A. 23
   B. 23 pairs
   C. 2

2. The sex chromosomes of a normal human male are:
   A. XX
   B. XY
   C. XXXX

3. The chromosomes that are not sex chromosomes are called:
   A. X and Y
   B. autosomes
   C. chromatin

4. In a pedigree, an affected male is shown by a:
   A. blank circle
   B. shaded circle
   C. shaded square

5. The inherited appearance is the:
   A. phenotype
   B. genotype
   C. co-dominance

6. Homozygous is known as:
   A. hybrid
   B. pure-breeding
   C. wild type

7. The less frequently inherited characteristic is called:
   A. recessive
   B. co-inherited
   C. dominant

8. Genes determining sex-linked conditions are found on the:
   A. autosomes
   B. Y chromosome
   C. X chromosome

9. A disease of blood clotting inability is:
   A. haemophilia
   B. haemochromatosis
   C. heparin

10. The ratio of phenotypes of offspring of a monohybrid cross is:
    A. 1:2:1
    B. 3:1
    C. 9:3:3:1

True-False Questions

11. Genes are made of DNA.
12. Somatic cells are body cells not gametes.
13. A variation of a gene is an allele.
14. Recessive characteristics are always indicated by capital letters.
15. If the genotype is brown eye colour, then the phenotype is BB.

Short Answer Questions

16. How can one determine the genotype of an animal with a dominant characteristic (e.g. a rabbit with brown fur)?
17. Why does the father determine the sex of a child?
18. How can two brown-eyed parents have a blue-eyed child?
19. Why do more males than females have sex-linked diseases such as red-green colour blindness?
20. What are the possible children's blood type if parents have A and AB blood types?

DIAGNOSTIC TEST
GENETICS
ANSWERS

1.B 2.B 3.B 4.C 5.A 6.B 7.A 8.C 9.A 10.B

11.T 12.T 13.T 14.F 15.F

16. Breed the animal with one which has the homozygous recessive characteristic (e.g. an albino rabbit). If any offspring are albino, the parent was heterozygous/hybrid. If no offspring ar albino, the parent is homozygous/pure-bred.
17. The mother has only the sex chromosome, the X chromosome, in the egg. If the father's sperm contains another X chromosome, the child will be a girl. If the sperm has a Y chromosome, the child will be a boy.
18. The allele for blue eye colour is recessive. Hence, two heterozygous/hybrid may carry the dominant brown allele as well as the recessive blue eye allele. If the two recessive alleles are in sperm and egg, the child will have blue eyes.
19. The sex-linked diseases are carried on the X chromosome. A male only has to have one X chromosome with the defective gene to inherit a sex-linked disease, but a female must have both chromosomes with defective genes to suffer from the disease. The latter is less likely.
20. If parents' genotypes are AA and AB – Possible child's genotypes are A and AB – Child could be A or AB type
    If parents' genotypes are AO and AB – Possible child's genotypes are AA, AB, AO, BO – Child could be A, B or AB types